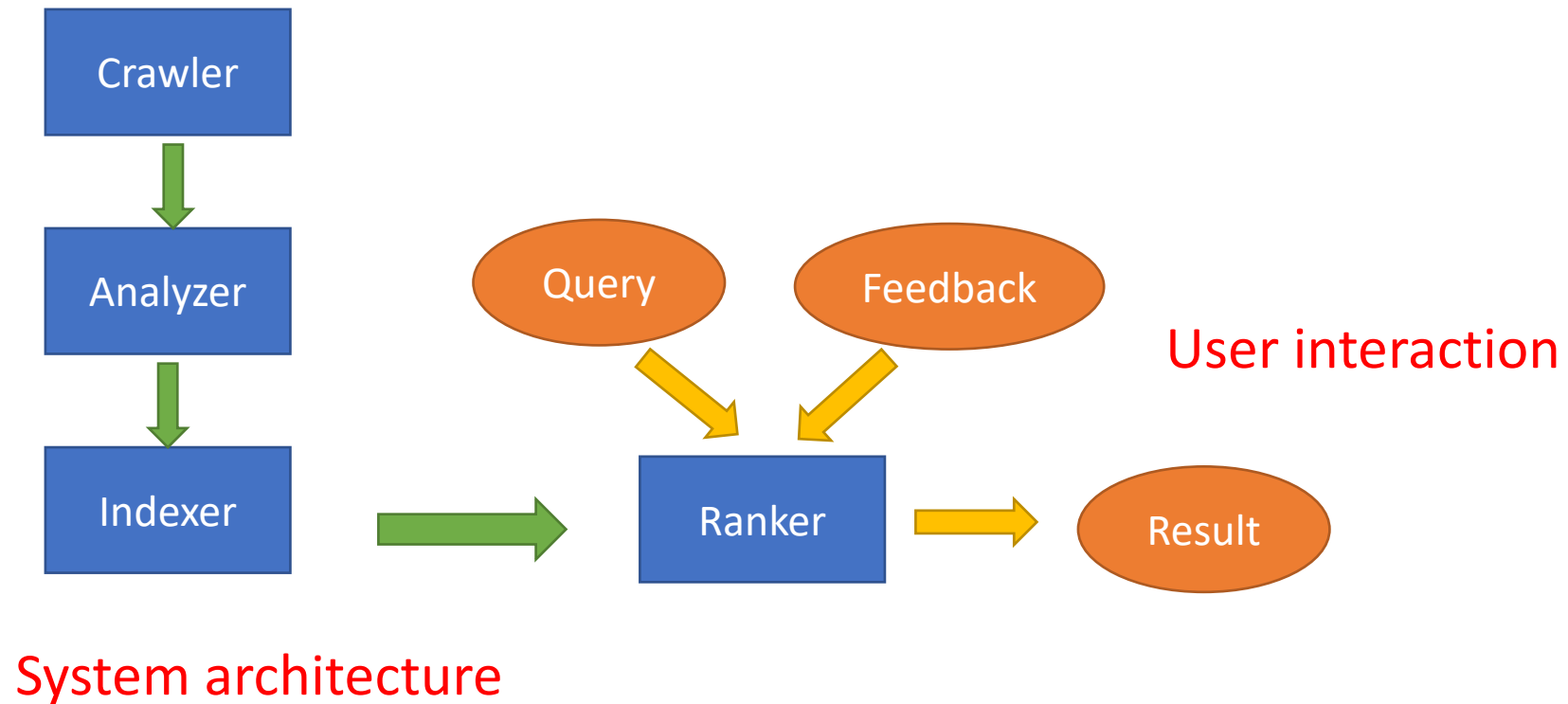


Lecture 4: Information Retrieval Evaluation

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Recap: Search engine architecture



Imagine

- You are the CTO of a e-commerce company and someone brings you a claimed state-of-the-art search engine named Googol. What should you do...
 - Redirect to Bing?
 - Try it yourself and see whether you like it?
- Maybe you want to investigate some details
 - How fast is the index?
 - How fast is the search/query?
 - Does it retrieve related goods?
- Key question: evaluate search quality
 - Satisfy users' information need

How to tell if information need is satisfied?

- Results relevant to query
 - How to assess?
- Results receive clicks
 - How about click bait/misleading information?
- Purchase the product
- 5-star reviews
- Repeat visitors/buyers
 - Threshold?

Quantify the IR quality measure

- Information need
 - *“an individual or group's desire to locate and obtain information to satisfy a conscious or unconscious need” – wiki*
 - Reflected by user query
 - Categorization of information need
 - Navigational
 - Informational
 - Transactional

Quantify the IR quality measure

- Satisfaction
 - *“the opinion of the user about a specific computer application, which they use” – wiki*
 - Reflected by
 - Increased result clicks
 - Repeated/increased visits
 - Result relevance

Satisfaction by measuring relevance



- Cranfield experiments
 - Pioneer work and foundation in IR evaluation
 - Basic hypothesis
 - Retrieved documents' relevance is a good proxy of a system's utility in satisfying users' information need
 - Procedure
 - 1,398 abstracts of aerodynamics journal articles
 - 225 queries
 - Exhaustive relevance judgments of all (query, document) pairs
 - Compare different indexing systems over such collection

Satisfaction by measuring relevance

- Three key elements for IR evaluation
 1. A document collection
 2. A test suite of information needs, expressible as queries
 3. A set of relevance judgments, e.g., binary assessment of either *relevant* or *non-relevant* for each query-document pair

Crowd-source relevance labels

- Present query-document pairs to low-cost labor on online crowd-sourcing platforms, e.g., Amazon Mechanical Turk
 - Cheaper than hiring qualified annotators
- Fair signal, but the variance of the labels can be high

Text REtrieval Conference (TREC)

- Large-scale evaluation of text retrieval methodologies
 - Since 1992, hosted by NIST
 - Standard benchmark for IR studies
 - A wide variety of evaluation collections
 - Web track
 - Question answering track
 - Cross-language track
 - Microblog track
 - And more...

TABLE 4.3 Common Test Corpora

<i>Collection</i>	<i>NDocs</i>	<i>NQrys</i>	<i>Size (MB)</i>	<i>Term/Doc</i>	<i>Q-D RelAss</i>
ADI	82	35			
ATT	2109	14	2	400	>10,000
CACM	3204	64	2	24.5	
CISI	1460	112	2	46.5	
Cranfield	1400	225	2	53.1	
LISA	5872	35	3		
Medline	1033	30	1		
NPL	11,429	93	3		
OSHMED	34,8566	106	400	250	16,140
Reuters	21,578	672	28	131	
TREC	740,000	200	2000	89-3543	» 100,000

Evaluation metric

- Given relevance judgements/labels, we still need to answer
 - Shall we perform stemming or stopword removal?
 - Which ranking method is the most effective?
- We need a quantifiable metric, by which we can compare different IR systems
 - As unranked retrieval sets
 - As ranked retrieval results

Unranked retrieval evaluation: Precision and Recall

- In a Boolean retrieval system
 - Precision: fraction of retrieved documents that are relevant, i.e., $p(\text{relevant}|\text{retrieved})$
 - Recall: fraction of relevant documents that are retrieved, i.e., $p(\text{retrieved}|\text{relevant})$

	relevant	nonrelevant
retrieved	true positive (TP)	false positive (FP)
not retrieved	false negative (FN)	true negative (TN)

Precision:

$$P = \frac{TP}{TP + FP}$$

Recall: $R = \frac{TP}{TP + FN}$

Unranked retrieval evaluation: Precision and Recall

- Precision and recall trade off against each other
- Precision decreases as the number of retrieved documents increases (unless in perfect ranking), while recall keeps increasing
- These two metrics emphasize different perspectives of an IR system
 - Precision: prefers systems retrieving fewer documents, but highly relevant
 - Recall: prefers systems retrieving more documents

F1 score

- Summarizing precision and recall to a single value
 - In order to compare different systems
 - F-measure: weighted harmonic mean of precision and recall, α balances the trade-off. E.g., F1 score.

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}} \quad \left(F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} \right)$$

- Why harmonic mean?

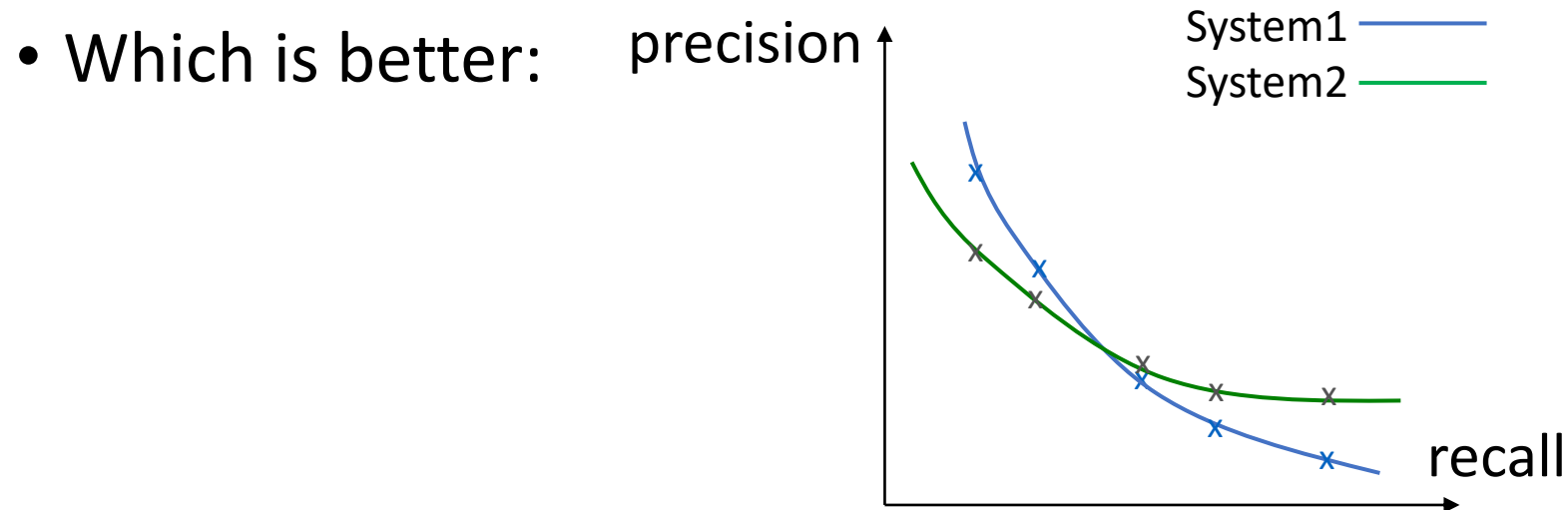
- System1: P:0.53, R:0.36
- System2: P:0.01, R:0.99

H	A
0.429	0.445
0.019	0.500

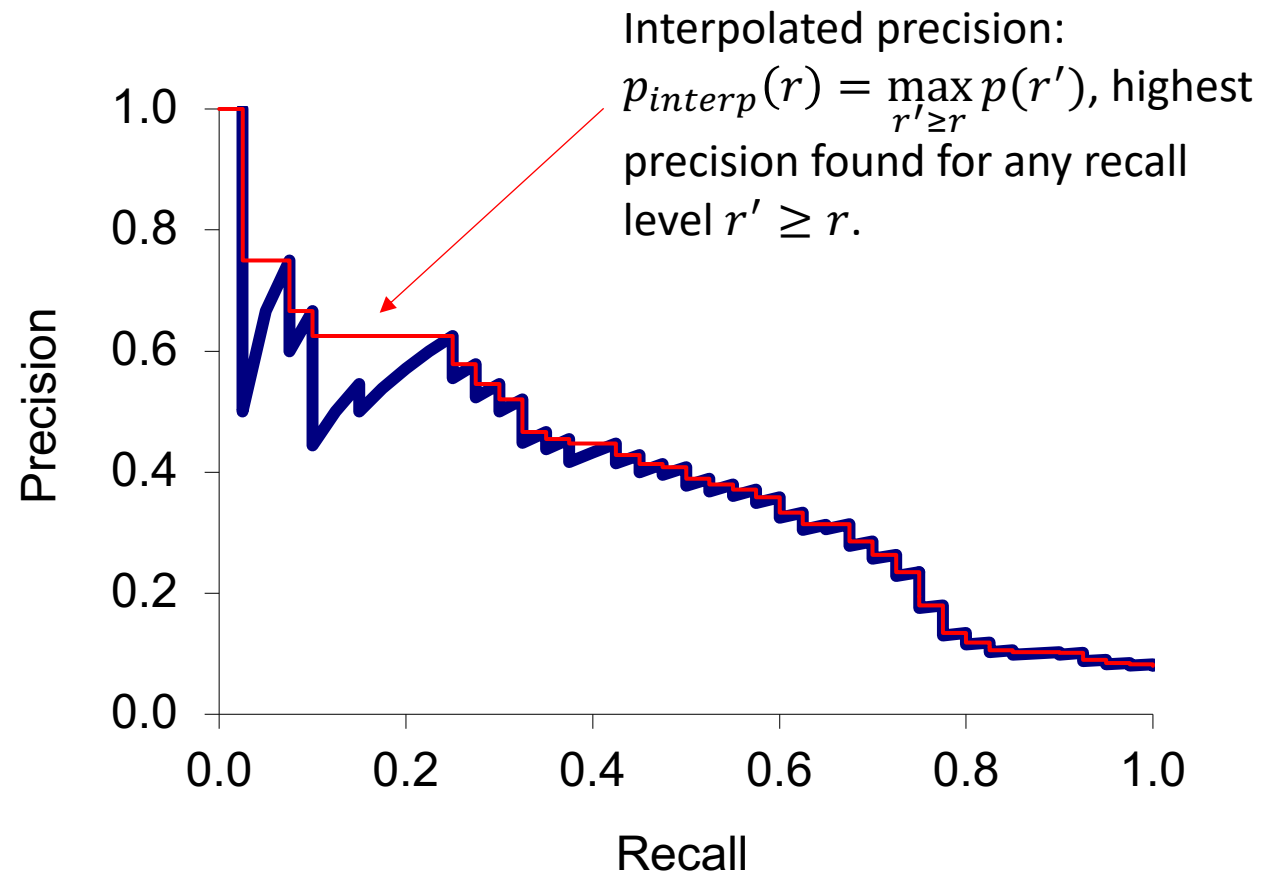
*Equal weight between
precision and recall*

Evaluation of ranked retrieval results

- Ranked results are the core feature of an IR system
 - Precision, recall and F-measure are set-based measures, which cannot assess the ranking quality
 - Solution: evaluate precision at every recall point



A precision-recall curve

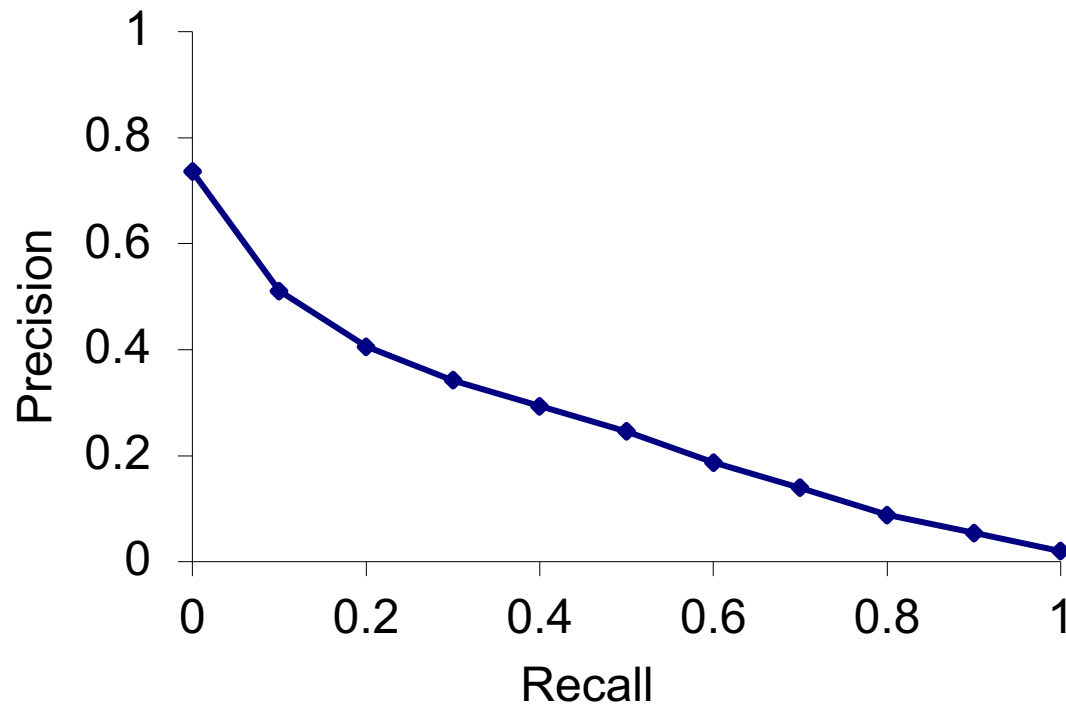


Evaluation of ranked retrieval results

- Summarize the ranking performance with a single number
- Binary relevance
 - Eleven-point interpolated average precision
 - Precision@K (P@K)
 - Mean Average Precision (MAP)
 - Mean Reciprocal Rank (MRR)
- Multiple grades of relevance
 - Normalized Discounted Cumulative Gain (NDCG)

Eleven-point interpolated average precision

- At the 11 recall levels $[0, 0.1, 0.2, \dots, 1.0]$, compute arithmetic mean of interpolated precision over all the queries



Precision@K

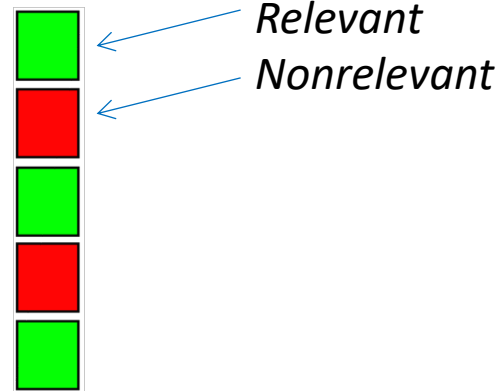
- Set a ranking position threshold K
- Ignores all documents ranked lower than K
- Compute precision in these top K retrieved documents

- E.g.,

P@3 of 2/3

P@4 of 2/4

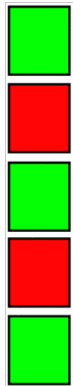
P@5 of 3/5



- In a similar fashion we have Recall@K

Mean Average Precision

- Consider rank position of each relevant doc
 - E.g., $K_1, K_2, \dots K_R$
- Compute $P@K$ for each $K_1, K_2, \dots K_R$
- Average precision = average of those $P@K$
 - E.g.,



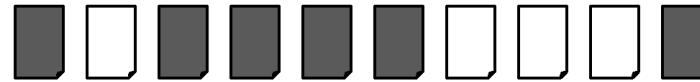
$$AvgPrec = \left(\frac{1}{1} + \frac{2}{3} + \frac{3}{5} \right) / 3$$

- MAP is the mean of Average Precision across multiple queries/rankings

AvgPrec is about one query

 = the relevant documents

Ranking #1



Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2



Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

AvgPrec of the two rankings

$$\text{Ranking \#1: } (1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$$

$$\text{Ranking \#2: } (0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$$

MAP is about a system

 = relevant documents for query 1

Ranking #1

										
Recall	0.2	0.2	0.4	0.4	0.4	0.6	0.6	0.6	0.8	1.0
Precision	1.0	0.5	0.67	0.5	0.4	0.5	0.43	0.38	0.44	0.5

 = relevant documents for query 2

Ranking #2

										
Recall	0.0	0.33	0.33	0.33	0.67	0.67	1.0	1.0	1.0	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.33	0.43	0.38	0.33	0.3

Query 1, $\text{AvgPrec} = (1.0 + 0.67 + 0.5 + 0.44 + 0.5) / 5 = 0.62$

Query 2, $\text{AvgPrec} = (0.5 + 0.4 + 0.43) / 3 = 0.44$

$\text{MAP} = (0.62 + 0.44) / 2 = 0.53$

MAP metric

- If a relevant document never gets retrieved, we assume the precision corresponding to that relevant document to be zero
- MAP is macro-averaging: each query counts equally
- MAP assumes users are interested in finding many relevant documents for each query
- MAP requires many relevance judgments in a text collection

Mean Reciprocal Rank (MRR)

- Measure the effectiveness of the ranked results
 - Suppose users are only looking for one relevant document
 - looking for a fact
 - known-item search
 - navigational queries
 - query auto completion
- Search duration $\sim$ Rank of the answer
 - Measures a user's effort

Mean Reciprocal Rank (MRR)

- Consider the rank position, K , of the first relevant document
- Reciprocal Rank = $\frac{1}{K}$
- MRR is the mean RR across multiple queries

Beyond binary relevance

- The level of documents' relevance quality with respect to a given query varies
- Highly relevant documents are more useful than marginally relevant documents
- The lower the ranked position of a relevant document is, the less useful it is for the user, since it is less likely to be examined
- ***Discounted Cumulative Gain***

Discounted Cumulative Gain

- Uses graded relevance as a measure of usefulness, or gain, from examining a document
- Gain is accumulated starting at the top of the ranking and discounted at lower ranks
- Typical discount is $1/\log(\text{rank})$
 - With base 2, the discount at rank 4 is $1/2$, and at rank 8 it is $1/3$

Discounted Cumulative Gain

- DCG is the total gain accumulated at a particular rank position p :

$$DCG_p = rel_1 + \sum_{i=2}^p \frac{rel_i}{\log_2 i}$$

 Relevance label at position i

- Alternative formulation

$$DCG_p = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(1 + i)}$$

- Standard metric in some web search companies
- Emphasize on retrieving highly relevant documents

Normalized Discounted Cumulative Gain

- Normalization is useful for contrasting queries with varying numbers of relevant results
- Normalize DCG at rank n by the DCG value at rank n of the ideal ranking
 - The ideal ranking is achieved via ranking documents with their relevance labels

NDCG - Example

5 documents: d_1, d_2, d_3, d_4, d_5

i	Ground Truth		Ranking Function ₁		Ranking Function ₂	
	Document Order	rel _i	Document Order	rel _i	Document Order	rel _i
1	d5	4	d3	2	d5	4
2	d4	3	d4	3	d3	2
3	d3	2	d2	1	d4	3
4	d2	1	d5	4	d1	0
5	d1	0	d1	0	d2	1

$$DCG_{GT} = \frac{2^4-1}{\log_2 2} + \frac{2^3-1}{\log_2 3} + \frac{2^2-1}{\log_2 4} + \frac{2^1-1}{\log_2 5} + \frac{2^0-1}{\log_2 6} = 21.35$$

$$DCG_{RF1} = \frac{2^2-1}{\log_2 2} + \frac{2^3-1}{\log_2 3} + \frac{2^1-1}{\log_2 4} + \frac{2^4-1}{\log_2 5} + \frac{2^0-1}{\log_2 6} = 14.38$$

$$DCG_{RF2} = \frac{2^4-1}{\log_2 2} + \frac{2^2-1}{\log_2 3} + \frac{2^3-1}{\log_2 4} + \frac{2^0-1}{\log_2 5} + \frac{2^1-1}{\log_2 6} = 20.78$$

NDCG - Example

5 documents: d_1, d_2, d_3, d_4, d_5

i	Ground Truth		Ranking Function ₁		Ranking Function ₂	
	Document Order	rel _i	Document Order	rel _i	Document Order	rel _i
1	d5	4	d3	2	d5	4
2	d4	3	d4	3	d3	2
3	d3	2	d2	1	d4	3
4	d2	1	d5	4	d1	0
5	d1	0	d1	0	d2	1
NDCG _{GT} =1.00		NDCG _{RF1} =0.67		NDCG _{RF2} =0.97		

$$DCG_{GT} = \frac{2^4-1}{\log_2 2} + \frac{2^3-1}{\log_2 3} + \frac{2^2-1}{\log_2 4} + \frac{2^1-1}{\log_2 5} + \frac{2^0-1}{\log_2 6} = 21.35$$

$$DCG_{RF1} = \frac{2^2-1}{\log_2 2} + \frac{2^3-1}{\log_2 3} + \frac{2^1-1}{\log_2 4} + \frac{2^4-1}{\log_2 5} + \frac{2^0-1}{\log_2 6} = 14.38$$

$$DCG_{RF2} = \frac{2^4-1}{\log_2 2} + \frac{2^2-1}{\log_2 3} + \frac{2^3-1}{\log_2 4} + \frac{2^0-1}{\log_2 5} + \frac{2^1-1}{\log_2 6} = 20.78$$

- *How about P@4, P@5, MAP and MRR?*

Quiz

Relevant documents: {A, B, C, D}

Result ranking:

A
D
E
G
F
C
H

P@5, AP, RR, NDCG?

Statistical significance tests

- How confident you are that an observed difference doesn't simply result from the particular queries you chose?

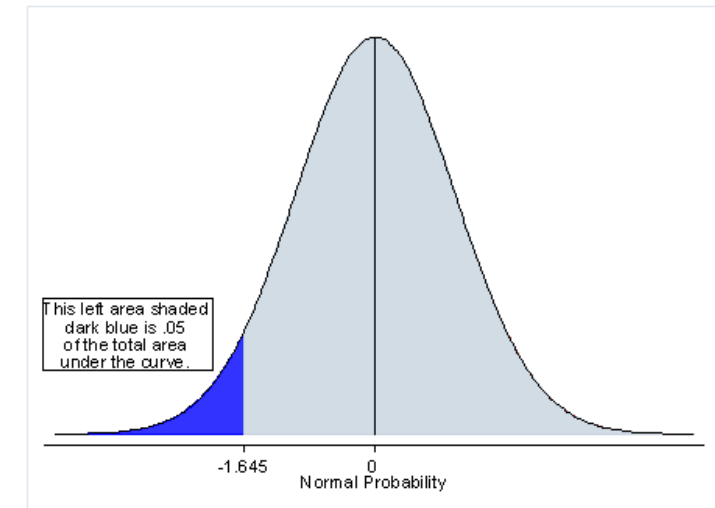
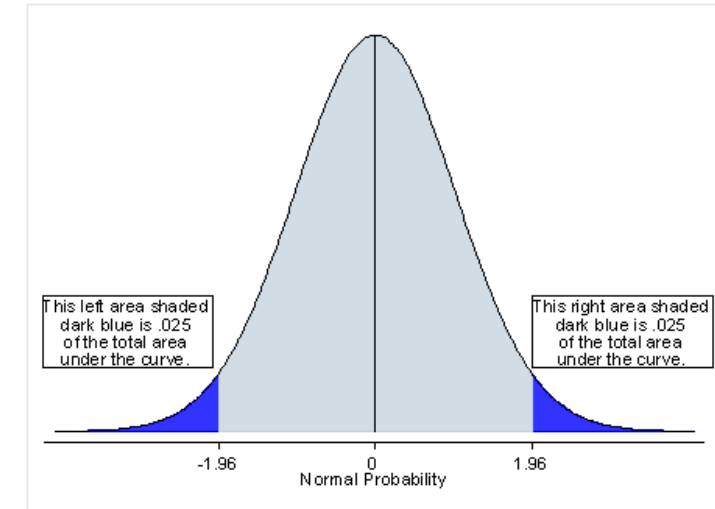
Experiment 1			Experiment 2		
<u>Query</u>	<u>System A</u>	<u>System B</u>	<u>Query</u>	<u>System A</u>	<u>System B</u>
1	0.20	0.40	11	0.02	0.76
2	0.21	0.41	12	0.39	0.07
3	0.22	0.42	13	0.26	0.17
4	0.19	0.39	14	0.38	0.31
5	0.17	0.37	15	0.14	0.02
6	0.20	0.40	16	0.09	0.91
7	0.21	0.41	17	0.12	0.56
Average	0.20	0.40	Average	0.20	0.40

Basic Statistics

- p -value in statistic test is the probability of obtaining data as extreme as was observed, if the null hypothesis was true (e.g., if observation is totally random)
- If p -value is smaller than the chosen significance level (α), we reject the null hypothesis (e.g., observation is not random)
- We seek to reject the null hypothesis (we seek to show that the observation is a random result), and so small p -values are good

Tests usually used in IR evaluations

- Sign test
 - Hypothesis: the difference median is zero between samples from two continuous distributions
- Wilcoxon signed rank test
 - Hypothesis: data are paired and come from the same population
- Paired t -test
 - Hypothesis: difference between two responses measured on the same statistical unit has a zero mean value
- One-tail v.s. two-tail?
 - If you aren't sure, use two-tail



Sign test

<u>Query</u>	<u>System A</u>	<u>System B</u>	<u>Sign Test</u>
11	0.02	0.76	+
12	0.39	0.07	-
13	0.26	0.17	-
14	0.38	0.31	-
15	0.14	0.02	-
16	0.09	0.91	+
17	0.12	0.56	+
Average	0.20	0.40	$p=0.7054$

Assumptions:

- 1) Comparisons are iid;
- 2) Comparisons are ordinal.

$H_0: W \sim B(m, 0.5)$, where
 W is the number of + sign.
 H_1 : A tends to be better or
B tends to be better.

Wilcoxon signed test

<u>Query</u>	<u>System A</u>	<u>System B</u>	<u>Wilcoxon Test</u>	
11	0.02	0.76	+ 6	Assumptions: 1) Comparisons are iid; 2) Comparisons are ordinal.
12	0.39	0.07	- 4	
13	0.26	0.17	- 2	<i>H_0: medians of the two samples are identical.</i>
14	0.38	0.31	- 1	
15	0.14	0.02	- 3	
16	0.09	0.91	+ 7	Sum of positive ranks: 18 Sum of negative ranks: 10 Critical value at N=7 is 3
17	0.12	0.56	+ 5	
Average	0.20	0.40		

Paired t-test

<u>Query</u>	<u>System A</u>	<u>System B</u>	<u>Paired t-test</u>	
11	0.02	0.76	+0.74	Assumptions: 1) equal sample size and variance; <u>or</u> 2) equal sample size but different variances. <i>H_0: no difference in mean of the two sets.</i>
12	0.39	0.07	-0.32	
13	0.26	0.17	-0.09	
14	0.38	0.31	-0.07	
15	0.14	0.02	-0.12	
16	0.09	0.91	+0.82	
17	0.12	0.56	+0.44	
Average	0.20	0.40	$p=0.2927$	

Summary

- Criterion for IR evaluation
- IR metrics
- Statistical test

Readings

- [MRS] CH 8.